PHENIKAA UNIVERSITY
SCHOOL OF INFORMATION SYSTEMS

COURSE CSE703095
SOFTWARE REQUIREMENTS

SAMPLE REPORT – LAB 07

Validation, Verification & Change Management

Topic: Illustrative example of a completed report (reference only, do not copy)

Case study project: MedBook – Online Medical Appointment Booking System
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1 Initial Result: Dataset Bugs Found

```
test_every_fr_has_a_valid_usecase ... FAIL
  AssertionError: FR-011 references a non-existent use case
test_high_priority_fr_must_have_testcase ... FAIL
  AssertionError: High-priority FRs with NO test case: ['FR-002', 'FR
     -007']
----------------------------------------------------------------------

Ran 7 tests in 0.002s
FAILED (failures=2)
```

Listing 1: unittest -v (before fixing)

2 Root Cause & Fix

Bug	Root cause	Fix
FR-011 → UC-11	Typo in the use case ID during specification in Lab 3 (UC-11 never existed; only UC-01..UC-10 are defined in `use_cases.csv`)	Either correct FR-011's `RelatedUseCase` field, or (as the team chose) create a new UC-11 "Verify insurance coverage" use case and add it to `use_cases.csv`
FR-002, FR-007 missing TC	When writing test cases in Labs 3/4, the team prioritized prominent "happy path" flows (booking/cancellation) and missed 2 less-visible flows	Added TC-011, TC-012 (Section 3)

3 Added Test Cases

ID	Requirement	Test case name
TC-011	FR-002	Doctor search returns the correct list filtered by specialty
TC-012	FR-007	Doctor updates visit results, fully logged with a timestamp

4 New Rule for Change Requests

```python
def test_approved_cr_has_estimated_impact(self):
    """Every 'Approved' CR must have a non-empty EstimatedImpact
       field."""
    crs = load("change_requests")
    missing = [c["ChangeRequestID"] for c in crs
               if c["Status"] == "Approved" and not c["
                  EstimatedImpact"].strip()]
    self.assertEqual(missing, [], f"Approved CRs missing
       EstimatedImpact: {missing}")
```

5 Result After Fixing

```
Ran 8 tests in 0.003s
OK
```

Listing 2: unittest -v (after fixing)

6 Impact Analysis of a Self-authored Change Request

CR-006: ”Add biometric login confirmation (fingerprint/Face ID) to the mobile app.” Using the RTM: directly affects FR-001 (Register/Login), indirectly affects NFR-003 (Security). Does not affect UC/TC of other flows ⇒ low risk, **recommended for approval** in phase 2.

7 Self-assessment

Criterion	Max	Self-score	Notes
Complete, accurate RTM	2.0	2.0	
Dataset fixed, tests PASS	3.0	3.0	8/8 PASS
New test cases + new rule	2.5	2.5	
Self-authored CR + impact analysis	1.5	1.5	
Presentation	1.0	1.0	
Total	**10.0**	**10.0**	